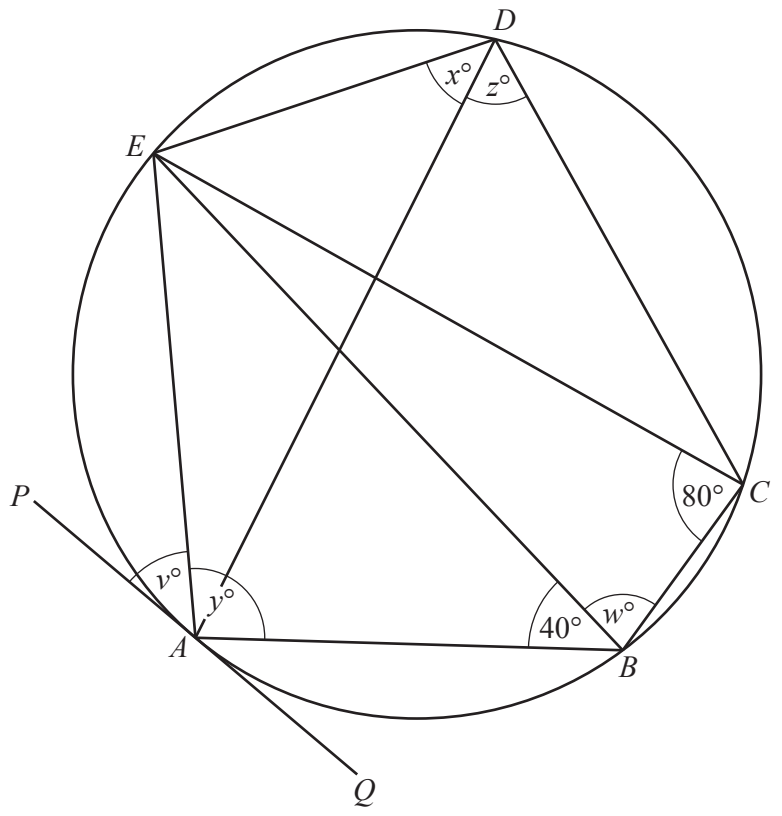
0580_w20_qp_42_Q8

8 (a)



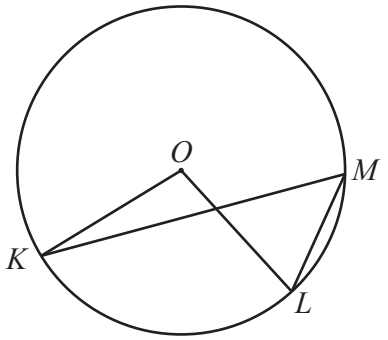
NOT TO SCALE

The points A, B, C, D and E lie on the circle.
PAQ is a tangent to the circle at A and $EC = EB$.
Angle $ECB = 80°$ and angle $ABE = 40°$.

Find the values of v, w, x, y and z.

$v =$ ..... $w =$ ..... $x =$ ..... $y =$ ..... $z =$ ..... [5]

(b)



NOT TO SCALE

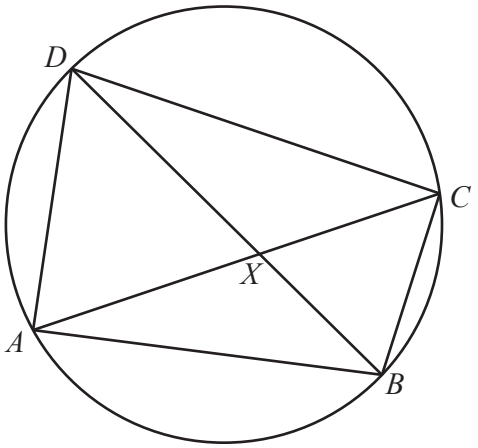
In the diagram, K, L and M lie on the circle, centre O.
Angle $KML = 2x°$ and reflex angle $KOL = 11x°$.

Find the value of x.

$x =$ ..... [3]

0580_w20_qp_42_Q8

(c)



NOT TO SCALE

The diagonals of the cyclic quadrilateral *ABCD* intersect at *X*.

(i) Explain why triangle *ADX* is similar to triangle *BCX*.
Give a reason for each statement you make.

.....

.....

.....

..... [3]

(ii) $AD = 10$ cm, $BC = 8$ cm, $BX = 5$ cm and $CX = 7$ cm.

(a) Calculate *DX*.

DX = ... cm [2]

(b) Calculate angle *BXC*.

Angle *BXC* = ... [4]

0580_w20_ms_42_Q8

Question	Answer	Marks	Partial Marks
8(a)	[v =] 40 [w =] 80 [x =] 40 [y =] 100 [z =] 60	**5**	**B1** for each **FT** angle z as 140 – *their w*
8(b)	24	**3**	**M2** for $360 - 11x = 2 \times 2x$ oe or **M1** for $360 - 11x$ seen or obtuse angle $KOL = 2 \times 2x$ oe
8(c)(i)	angle ADX = angle BCX oe same segment oe angle DAX = angle CBX oe same segment oe angle AXD = BXC oe [vertically] opposite oe	**M2**	Accept in any order **M1** for one correct pair with reason If 0 scored, **SC1** for two correct pairs of equal angles identified with incorrect/no reasons
	corresponding angles are equal oe	**A1**	
8(c)(ii)(a)	8.75 or 8¾	**2**	**M1** for $\frac{8}{10} = \frac{7}{DX}$ oe
8(c)(ii)(b)	81.8 or 81.78 to 81.79	**4**	**M2** for [cos[BXC] =] $\frac{5^2 + 7^2 - 8^2}{2 \times 5 \times 7}$ oe or **M1** for $8^2 = 5^2 + 7^2 - 2 \times 5 \times 7 \times \cos(...)$ oe **A1** for $\frac{10}{70}$ oe